NeonPine – materials compliance (RoHS / REACH) data collection

The RoHS assessment follows EN IEC 63000: rate each part's risk, then collect evidence proportional to that risk. "Evidence" is a supplier RoHS/REACH declaration, a full material declaration (IPC-1752A), or a test report.

BOM line	Part	Supplier	Risk	Evidence needed	Status
LA1–LA10	Brightmaster NEON-3 neon lamp	TME / Brightmaster	**High** (glass, lead-in wires, small brand, possible Kr-85)	RoHS + REACH declaration **and a written statement "no radioactive substances"**; for the first lot, an XRF check of the leads	open
U1	CH32V003F4P6	WCH	Low	RoHS/REACH declaration from WCH	open
Q1	DMN15H310SE	Diodes Inc.	Low	datasheet says RoHS, halogen/antimony-free; keep the datasheet	ok (datasheet)
Q11–Q20	MMBTA42	`[make]`	Low	manufacturer declaration	open
D1, D2	ES1D, SMAJ100A	`[make]`	Low	declaration	open
L11	SWPA6028S331MT 330 µH	Sunlord	Medium (ferrite, magnet wire)	declaration	open
C1–C7	MLCC	`[make]`	Low	declaration	open
R*	thick-film chip resistors	`[make]`	Low (glass-frit Pb exempt 7(c)-I)	declaration, exemptions listed	open
~~R8~~ → **Q2**	~~GL5528 CdS LDR~~ → **Vishay TEPT4400** phototransistor	Vishay	Low	Vishay RoHS / REACH declaration (Cd-free silicon)	**rev B: replaced** – declaration to file
J1	USB-C GCT USB4125-GF-A	GCT	Medium (copper alloy Pb, exemption 6(c))	declaration incl. exemptions → Prop 65 decision	open
F1	MF-NSMF050-2 PTC 1206	Bourns	Low	declaration	open
D3 / FB1	SMF5.0A TVS / 600 Ω ferrite 0603	`[make]`	Low	declaration	open

BOM line	Part	Supplier	Risk	Evidence needed	Status
J2	~~pin header~~ → bare pogo pads (rev B)	–	–	none (no part)	**removed**
mech	rear cover + stand PCBs; 3 × M2 × 5 standoff, 6 × M2 × 4 screw	`[fab]`, `[make]`	Medium (brass standoffs: Pb exemption 6(c))	fab statement; standoff/screw declarations	open
PCB	FR-4, ENIG, LPI mask, silkscreen	`[fab]`	Low	fab RoHS/REACH statement, UL 94 V-0 file number	open
Solder	SAC305 paste + wire	`[assembler]`	Low	declaration	open
Packaging	carton, insert	`[supplier]`	Low	PPWR heavy-metal declaration	open

✔ **Resolved in rev B:** the CdS photoresistor GL5528 (cadmium, not RoHS-compliant as a standalone part) was replaced by the Vishay TEPT4400 silicon phototransistor (+5V → C, E → PD5, R9 = 22k).